

Understanding Overfitting

Deep Learning

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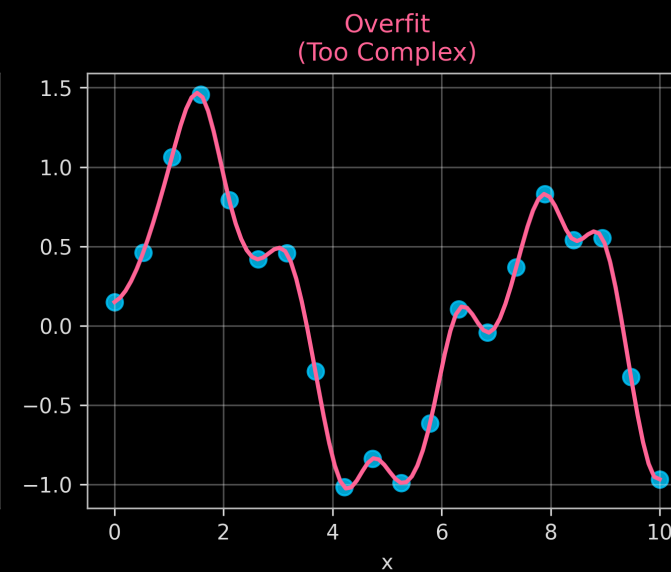
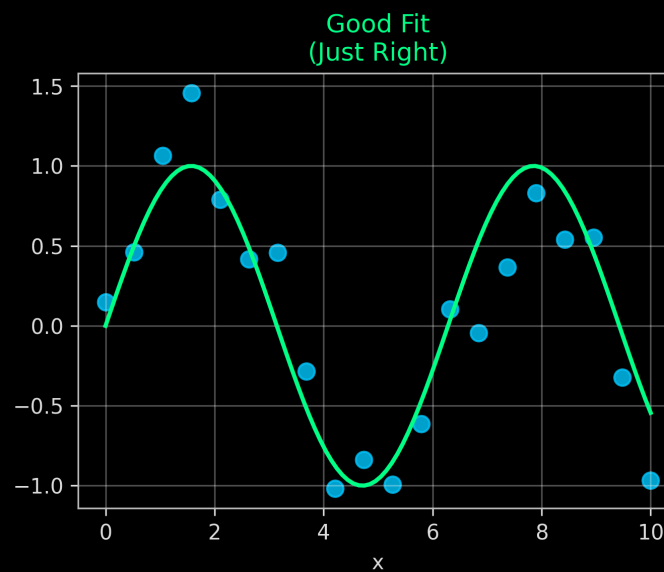
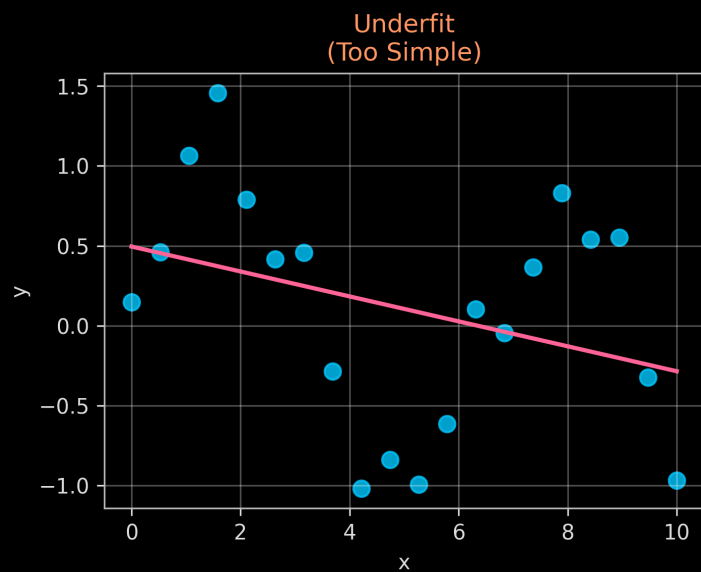
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Contents of This Video

In this video, we will cover:

- What is overfitting?
- Why neural networks overfit easily
- The bias-variance trade-off
- Recognizing overfitting from training curves
- Preview of solutions

What is Overfitting?



The Overfitting Problem

Overfitting = memorizing instead of learning

Metric	Good Model	Overfit Model
Training Accuracy	90%	100%
Test Accuracy	88%	60%

Perfect training performance $\neq$ good model!

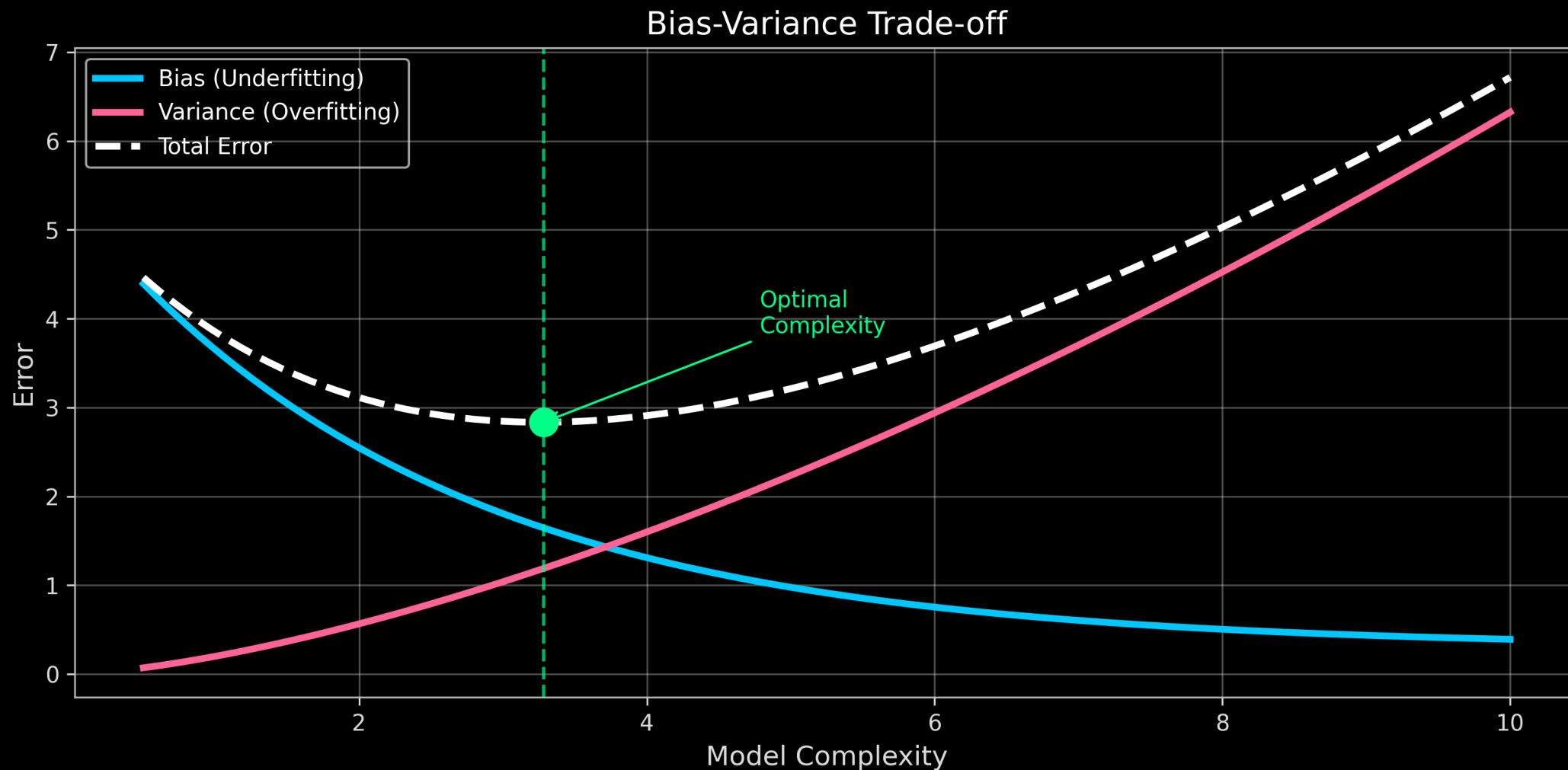
Why Neural Networks Overfit

Many parameters = high capacity

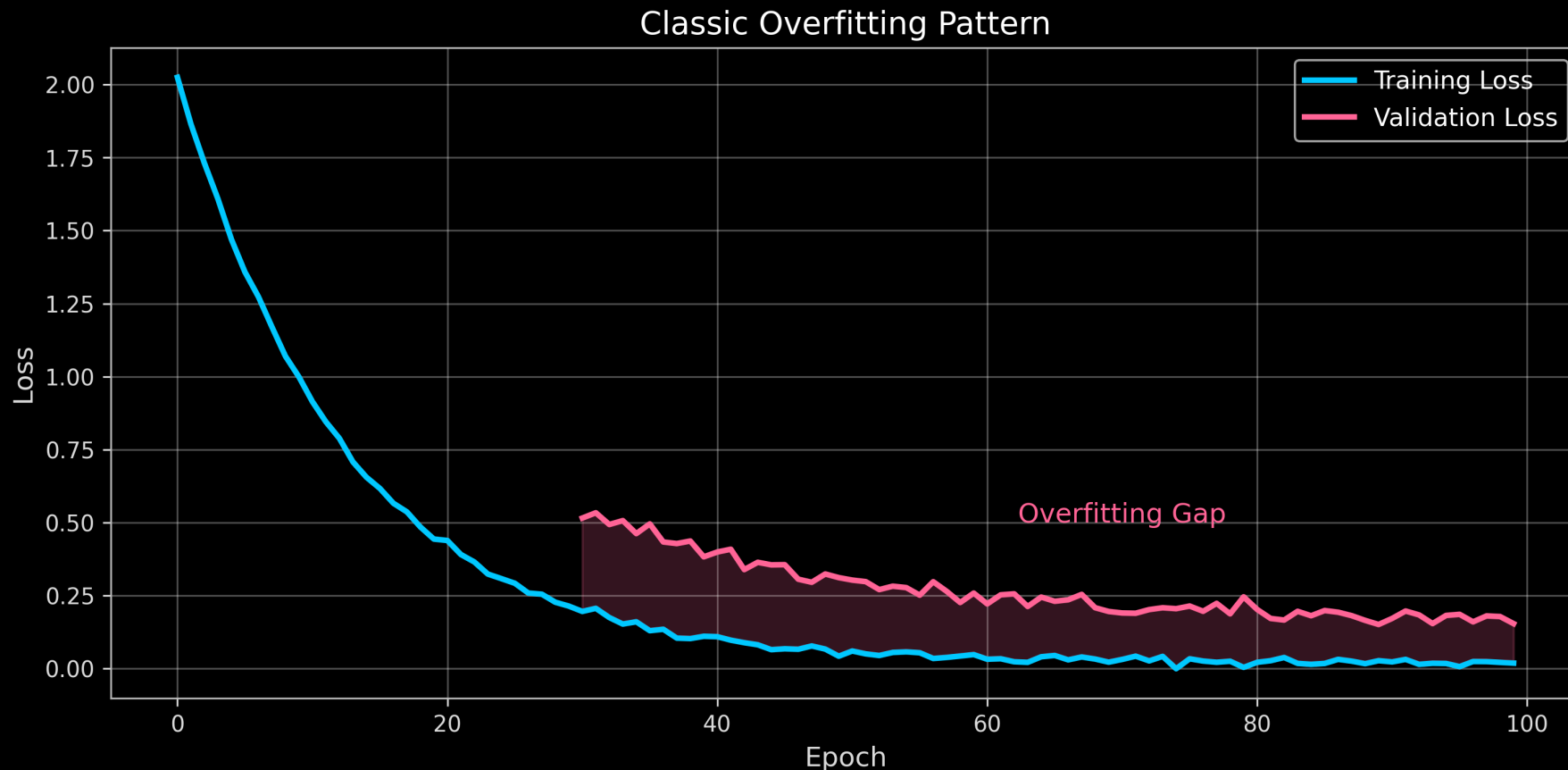
Network	Parameters
Small MLP (784→64→10)	~50,000
ResNet-50	~25,000,000
GPT-3	~175,000,000,000

A network with more parameters than training samples can memorize everything!

The Bias-Variance Trade-off



Recognizing Overfitting



Warning signs:

- Train accuracy >> Validation accuracy
- Validation loss increases while train decreases

Solutions Preview

Regularization techniques (coming next!):

- **L2 Regularization (Weight Decay):** Penalize large weights
- **Dropout:** Randomly drop neurons during training
- **Early Stopping:** Stop before overfitting occurs
- **Data Augmentation:** Artificially expand training data
- **Batch Normalization:** Stabilize and regularize

The goal: Allow high-capacity models while preventing memorization

What We've Covered

In this video, we've learned:

- Overfitting = model memorizes training data instead of learning patterns
- Neural networks overfit easily due to high parameter counts
- Bias-variance trade-off: complexity vs. generalization
- Recognize overfitting: train/val gap grows, val loss increases
- Solution: regularization techniques (covered next)